

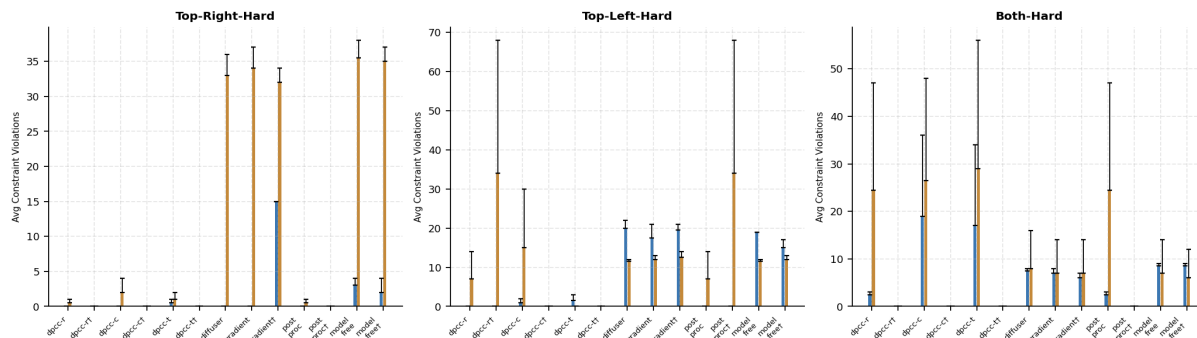
Diffusion vs Flow Matching

Full Data Analysis — avoiding-d3il Benchmark · Seed 6 · All Scenarios & Methods

A = Diffusion (GaussianDiffusion,
logs/avoiding-d3il/diffusion/)

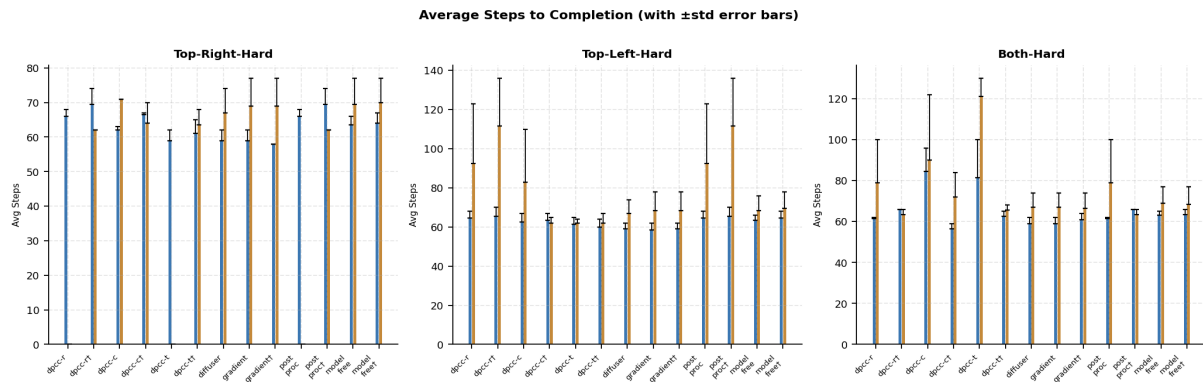
B = Flow Matching (FM,
logs/avoiding-d3il/flow_matching/)

This report presents the complete raw data from both evaluation logs. Three scenarios are tested: **top-right-hard**, **top-left-hard**, and **both-hard**. Seventeen methods are evaluated per scenario, including core DPCC variants, baselines, and dt-sensitivity ablations of dpcc-c-tightened. All six metrics reported in the raw logs are tabulated: success rate, constraint satisfaction, goal+constraint success rate, average steps, average constraint violations, average total violation magnitude, and computation time per step.



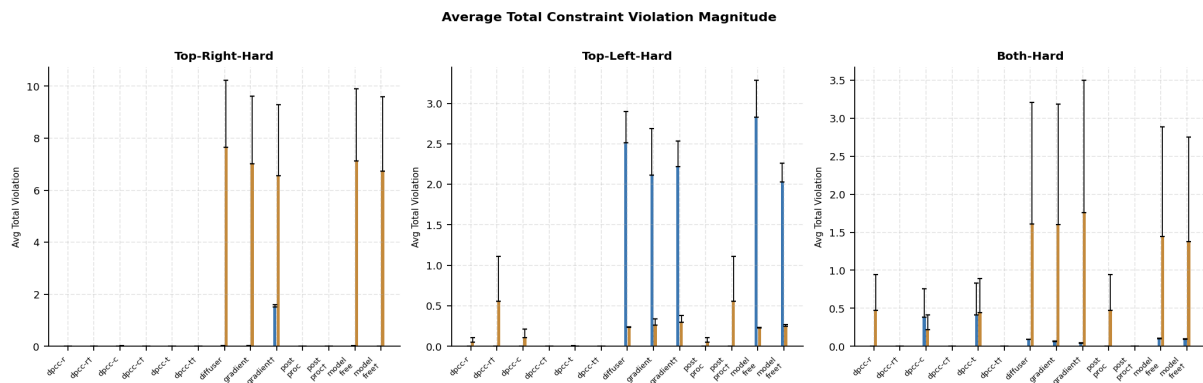
1.3 Average Steps to Completion

Fewer steps = more efficient trajectory. Methods that fail (0 steps) indicate the agent never reached the goal. Error bars = $\pm$ std.



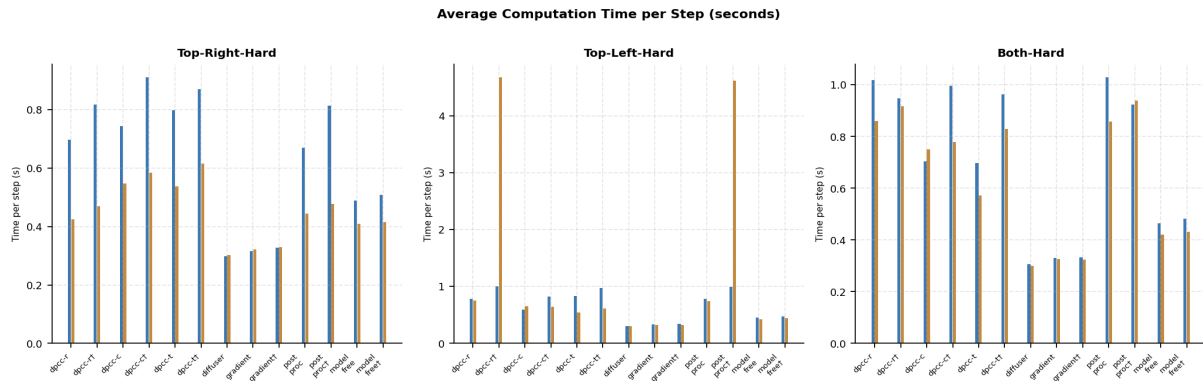
1.4 Average Total Constraint Violation Magnitude

Cumulative violation magnitude over an episode (not just count). Even when violation count is low, large magnitudes indicate severe constraint breaches.



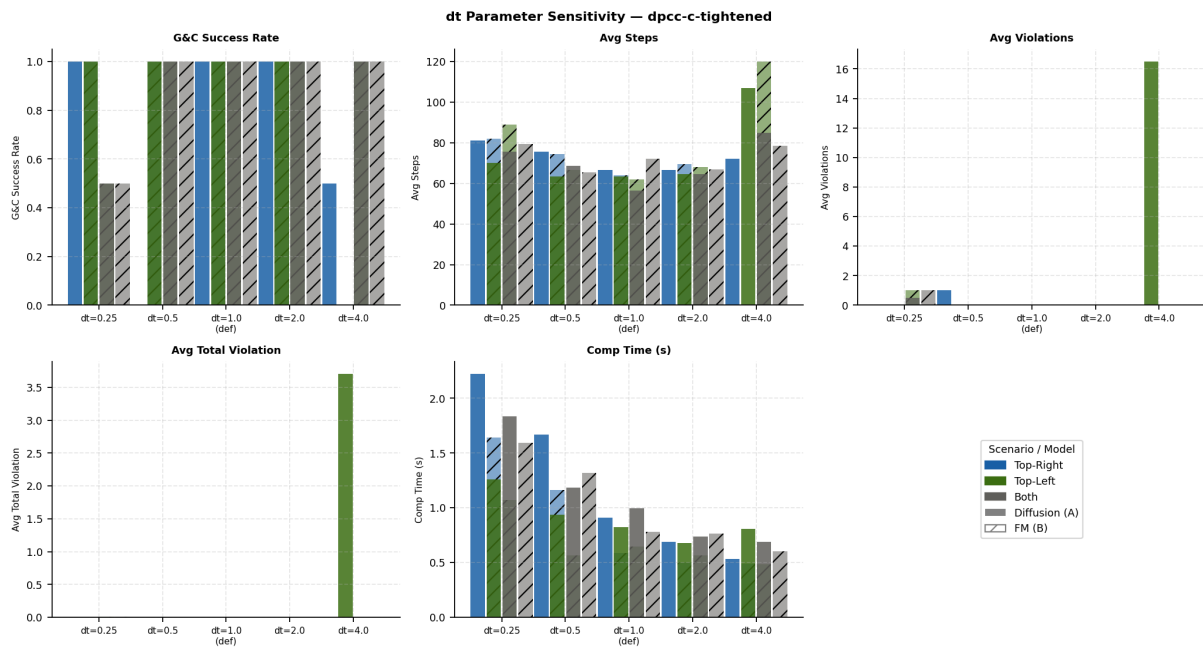
1.5 Computation Time per Step

Wall-clock seconds per planning step. DPCC methods with projection are slower than baselines. Post-processing and dpcc-r are among the fastest DPCC variants.



1.6 dt Parameter Sensitivity — dpcc-c-tightened

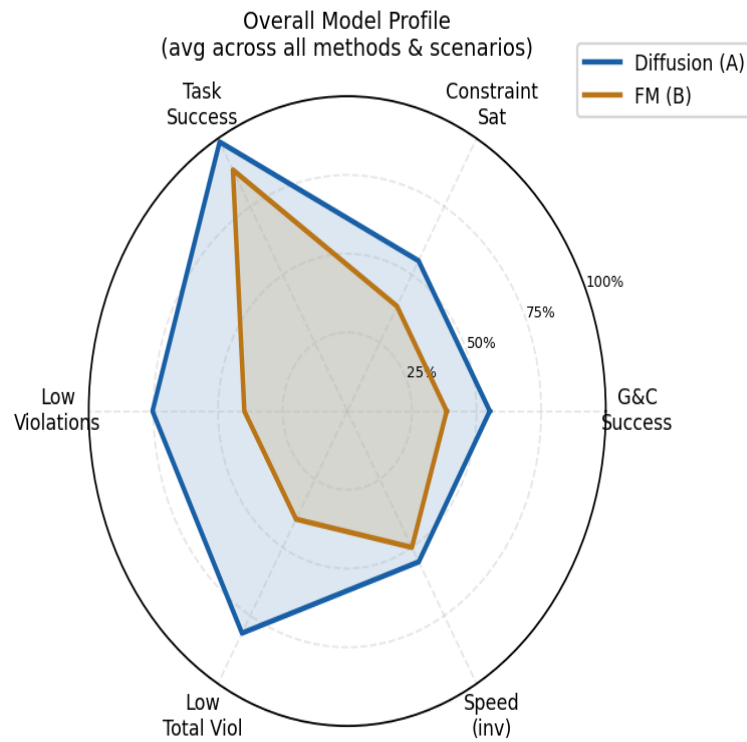
Ablation of the dt hyperparameter across all three scenarios. Solid bars = Diffusion (A), hatched = FM (B). Colors distinguish scenarios. dt=1.0 is the default configuration.



1.7 Overall Model Profile Radar

Aggregated across all 13 core methods and 3 scenarios. 'Low Violations' and 'Low Total Viol' are inverted (higher = fewer violations).

'Speed (inv)' = inverse of mean compute time.



Section 2 — Complete Raw Data Tables

Every metric from both raw log files is reproduced below. Tables are split by scenario. Column abbreviations:

SR=Success Rate, **CS**=Constraints Satisfied, **G&C**=Goal & Constraints, **Steps**=Avg Steps $\pm$ std, **Viol#**=Avg Violations $\pm$ std, **Viol Σ** =Avg Total Violation $\pm$ std, **Time**=Avg Comp Time/step (s), **TE**=Tracking Error.

2.1 Top-Right-Hard

Diffusion (A)

Method	SR	CS	G&C	Steps $\pm$ std	Viol# $\pm$ std	Viol Σ $\pm$ std	Time (s)	TE
dpcc-r	100%	100%	100%	66.0 $\pm$ 2.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.698	–
dpcc-r-tight	100%	100%	100%	69.5 $\pm$ 4.5	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.817	–
dpcc-c	100%	100%	100%	62.0 $\pm$ 1.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.744	–
dpcc-c-tight	100%	100%	100%	66.5 $\pm$ 0.5	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.911	–
dpcc-t	100%	50%	50%	59.0 $\pm$ 3.0	0.5 $\pm$ 0.5	0.000 $\pm$ 0.000	0.799	–
dpcc-t-tight	100%	100%	100%	61.0 $\pm$ 4.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.871	–
diffuser	100%	100%	100%	59.0 $\pm$ 3.0	0.0 $\pm$ 0.0	0.021 $\pm$ 0.002	0.298	0.025
gradient	100%	100%	100%	59.0 $\pm$ 3.0	0.0 $\pm$ 0.0	0.021 $\pm$ 0.002	0.316	–
gradient-tight	50%	0%	0%	58.0 $\pm$ 0.0	15.0 $\pm$ 0.0	1.521 $\pm$ 0.082	0.327	–
post_proc	100%	100%	100%	66.0 $\pm$ 2.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.670	–
post_proc-tight	100%	100%	100%	69.5 $\pm$ 4.5	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.814	–
model_free	100%	0%	0%	63.5 $\pm$ 2.5	3.0 $\pm$ 1.0	0.012 $\pm$ 0.011	0.490	–
model_free-tight	100%	50%	50%	64.0 $\pm$ 3.0	2.0 $\pm$ 2.0	0.009 $\pm$ 0.009	0.508	–
dpcc-c-tight-dt0.25	100%	100%	100%	81.0 $\pm$ 6.0	0.0 $\pm$ 0.0	0.002 $\pm$ 0.000	2.225	–
dpcc-c-tight-dt0.5	100%	0%	0%	75.5 $\pm$ 2.5	1.0 $\pm$ 0.0	0.003 $\pm$ 0.000	1.668	–
dpcc-c-tight-dt2.0	100%	100%	100%	66.5 $\pm$ 0.5	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.689	–
dpcc-c-tight-dt4.0	50%	50%	50%	72.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.535	–

FM (B)

Method	SR	CS	G&C	Steps $\pm$ std	Viol# $\pm$ std	Viol Σ $\pm$ std	Time (s)	TE
dpcc-r	0%	0%	0%	0.0 $\pm$ 0.0	0.5 $\pm$ 0.5	0.002 $\pm$ 0.002	0.424	–
dpcc-r-tight	50%	50%	50%	62.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.469	–
dpcc-c	50%	0%	0%	71.0 $\pm$ 0.0	2.0 $\pm$ 2.0	0.015 $\pm$ 0.015	0.547	–
dpcc-c-tight	100%	100%	100%	64.0 $\pm$ 6.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.585	–
dpcc-t	0%	0%	0%	0.0 $\pm$ 0.0	1.0 $\pm$ 1.0	0.010 $\pm$ 0.010	0.537	–
dpcc-t-tight	100%	100%	100%	63.5 $\pm$ 4.5	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.615	–
diffuser	100%	0%	0%	67.0 $\pm$ 7.0	33.0 $\pm$ 3.0	7.654 $\pm$ 2.573	0.303	0.035
gradient	100%	0%	0%	69.0 $\pm$ 8.0	34.0 $\pm$ 3.0	7.014 $\pm$ 2.598	0.321	–
gradient-tight	100%	0%	0%	69.0 $\pm$ 8.0	32.0 $\pm$ 2.0	6.556 $\pm$ 2.728	0.329	–
post_proc	0%	0%	0%	0.0 $\pm$ 0.0	0.5 $\pm$ 0.5	0.002 $\pm$ 0.002	0.444	–
post_proc-tight	50%	50%	50%	62.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.478	–
model_free	100%	0%	0%	69.5 $\pm$ 7.5	35.5 $\pm$ 2.5	7.118 $\pm$ 2.776	0.409	–

model_free-tight	100%	0%	0%	70.0 ±7.0	35.0 ±2.0	6.726 ±2.879	0.416	–
dpcc-c-tight-dt0.25	100%	100%	100%	82.0 ±5.0	0.0 ±0.0	0.003 ±0.002	1.644	–
dpcc-c-tight-dt0.5	100%	100%	100%	74.5 ±6.5	0.0 ±0.0	0.001 ±0.001	1.163	–
dpcc-c-tight-dt2.0	100%	100%	100%	69.5 ±8.5	0.0 ±0.0	0.000 ±0.000	0.500	–
dpcc-c-tight-dt4.0	0%	0%	0%	0.0 ±0.0	0.0 ±0.0	0.000 ±0.000	0.491	–

2.2 Top-Left-Hard

Diffusion (A)

Method	SR	CS	G&C;	Steps ± std	Viol# ± std	ViolΣ ± std	Time (s)	TE
dpcc-r	100%	100%	100%	64.5 ±3.5	0.0 ±0.0	0.000 ±0.000	0.784	–
dpcc-r-tight	100%	100%	100%	65.5 ±4.5	0.0 ±0.0	0.000 ±0.000	0.998	–
dpcc-c	100%	50%	50%	62.5 ±4.5	1.0 ±1.0	0.003 ±0.003	0.592	–
dpcc-c-tight	100%	100%	100%	63.5 ±3.5	0.0 ±0.0	0.000 ±0.000	0.822	–
dpcc-t	100%	50%	50%	61.5 ±3.5	1.5 ±1.5	0.004 ±0.004	0.829	–
dpcc-t-tight	100%	100%	100%	60.0 ±4.0	0.0 ±0.0	0.000 ±0.000	0.976	–
diffuser	100%	0%	0%	59.0 ±3.0	20.0 ±2.0	2.515 ±0.387	0.306	0.025
gradient	100%	0%	0%	58.5 ±3.5	17.5 ±3.5	2.116 ±0.571	0.334	–
gradient-tight	100%	0%	0%	59.0 ±3.0	19.5 ±1.5	2.218 ±0.316	0.348	–
post_proc	100%	100%	100%	64.5 ±3.5	0.0 ±0.0	0.000 ±0.000	0.782	–
post_proc-tight	100%	100%	100%	65.5 ±4.5	0.0 ±0.0	0.000 ±0.000	0.996	–
model_free	100%	0%	0%	63.5 ±2.5	19.0 ±0.0	2.829 ±0.455	0.458	–
model_free-tight	100%	0%	0%	64.5 ±3.5	15.0 ±2.0	2.029 ±0.233	0.470	–
dpcc-c-tight-dt0.25	100%	100%	100%	70.0 ±3.0	0.0 ±0.0	0.000 ±0.000	1.259	–
dpcc-c-tight-dt0.5	100%	100%	100%	63.5 ±0.5	0.0 ±0.0	0.000 ±0.000	0.937	–
dpcc-c-tight-dt2.0	100%	100%	100%	64.5 ±0.5	0.0 ±0.0	0.000 ±0.000	0.679	–
dpcc-c-tight-dt4.0	100%	0%	0%	107.0 ±2.0	16.5 ±0.5	3.711 ±0.053	0.810	–

FM (B)

Method	SR	CS	G&C;	Steps ± std	Viol# ± std	ViolΣ ± std	Time (s)	TE
dpcc-r	100%	50%	50%	92.5 ±30.5	7.0 ±7.0	0.055 ±0.055	0.753	–
dpcc-r-tight	100%	50%	50%	111.5 ±24.5	34.0 ±34.0	0.557 ±0.557	4.675	–
dpcc-c	100%	50%	50%	83.0 ±27.0	15.0 ±15.0	0.105 ±0.105	0.656	–
dpcc-c-tight	100%	100%	100%	62.0 ±3.0	0.0 ±0.0	0.000 ±0.000	0.646	–
dpcc-t	100%	100%	100%	62.0 ±2.0	0.0 ±0.0	0.000 ±0.000	0.548	–
dpcc-t-tight	100%	100%	100%	62.0 ±5.0	0.0 ±0.0	0.000 ±0.000	0.614	–
diffuser	100%	0%	0%	67.0 ±7.0	11.5 ±0.5	0.237 ±0.005	0.304	0.035
gradient	100%	0%	0%	68.5 ±9.5	12.0 ±1.0	0.263 ±0.077	0.326	–
gradient-tight	100%	0%	0%	68.5 ±9.5	12.5 ±1.5	0.299 ±0.085	0.324	–
post_proc	100%	50%	50%	92.5 ±30.5	7.0 ±7.0	0.055 ±0.055	0.744	–
post_proc-tight	100%	50%	50%	111.5 ±24.5	34.0 ±34.0	0.557 ±0.557	4.615	–

model_free	100%	0%	0%	68.5 ±7.5	11.5 ±0.5	0.225 ±0.008	0.422	–
model_free-tight	100%	0%	0%	69.5 ±8.5	12.0 ±1.0	0.245 ±0.028	0.441	–
dpcc-c-tight-dt0.25	100%	50%	50%	89.0 ±7.0	1.0 ±1.0	0.013 ±0.009	1.071	–
dpcc-c-tight-dt0.5	100%	100%	100%	66.5 ±0.5	0.0 ±0.0	0.000 ±0.000	0.563	–
dpcc-c-tight-dt2.0	100%	100%	100%	68.0 ±8.0	0.0 ±0.0	0.000 ±0.000	0.563	–
dpcc-c-tight-dt4.0	100%	100%	100%	120.0 ±6.0	0.0 ±0.0	0.000 ±0.000	0.486	–

2.3 Both-Hard

Diffusion (A)

Method	SR	CS	G&C;	Steps ± std	Viol# ± std	ViolΣ ± std	Time (s)	TE
dpcc-r	100%	0%	0%	61.5 ±0.5	2.5 ±0.5	0.002 ±0.001	1.019	–
dpcc-r-tight	100%	100%	100%	66.0 ±0.0	0.0 ±0.0	0.000 ±0.000	0.948	–
dpcc-c	100%	0%	0%	84.5 ±11.5	19.0 ±17.0	0.381 ±0.377	0.704	–
dpcc-c-tight	100%	100%	100%	56.5 ±2.5	0.0 ±0.0	0.000 ±0.000	0.997	–
dpcc-t	100%	50%	50%	81.5 ±18.5	17.0 ±17.0	0.416 ±0.416	0.697	–
dpcc-t-tight	100%	100%	100%	62.5 ±2.5	0.0 ±0.0	0.000 ±0.000	0.964	–
diffuser	100%	0%	0%	59.0 ±3.0	7.5 ±0.5	0.090 ±0.001	0.307	0.025
gradient	100%	0%	0%	59.0 ±3.0	7.0 ±1.0	0.064 ±0.007	0.331	–
gradient-tight	100%	0%	0%	61.0 ±3.0	6.0 ±1.0	0.044 ±0.004	0.332	–
post_proc	100%	0%	0%	61.5 ±0.5	2.5 ±0.5	0.002 ±0.001	1.029	–
post_proc-tight	100%	100%	100%	66.0 ±0.0	0.0 ±0.0	0.000 ±0.000	0.924	–
model_free	100%	0%	0%	63.0 ±2.0	8.5 ±0.5	0.103 ±0.008	0.465	–
model_free-tight	100%	0%	0%	63.5 ±2.5	8.5 ±0.5	0.091 ±0.010	0.481	–
dpcc-c-tight-dt0.25	100%	50%	50%	75.5 ±0.5	0.5 ±0.5	0.000 ±0.000	1.835	–
dpcc-c-tight-dt0.5	100%	100%	100%	68.5 ±4.5	0.0 ±0.0	0.000 ±0.000	1.185	–
dpcc-c-tight-dt2.0	100%	100%	100%	64.5 ±2.5	0.0 ±0.0	0.000 ±0.000	0.737	–
dpcc-c-tight-dt4.0	100%	100%	100%	85.0 ±2.0	0.0 ±0.0	0.000 ±0.000	0.691	–

FM (B)

Method	SR	CS	G&C;	Steps ± std	Viol# ± std	ViolΣ ± std	Time (s)	TE
dpcc-r	100%	0%	0%	79.0 ±21.0	24.5 ±22.5	0.473 ±0.469	0.859	–
dpcc-r-tight	100%	100%	100%	63.5 ±2.5	0.0 ±0.0	0.000 ±0.000	0.916	–
dpcc-c	100%	0%	0%	90.0 ±32.0	26.5 ±21.5	0.217 ±0.197	0.751	–
dpcc-c-tight	100%	100%	100%	72.0 ±12.0	0.0 ±0.0	0.000 ±0.000	0.779	–
dpcc-t	100%	0%	0%	121.0 ±9.0	29.0 ±27.0	0.445 ±0.444	0.573	–
dpcc-t-tight	100%	100%	100%	65.5 ±2.5	0.0 ±0.0	0.000 ±0.000	0.830	–
diffuser	100%	50%	50%	67.0 ±7.0	8.0 ±8.0	1.612 ±1.594	0.300	0.035
gradient	100%	50%	50%	67.0 ±7.0	7.0 ±7.0	1.602 ±1.584	0.326	–
gradient-tight	100%	50%	50%	66.5 ±7.5	7.0 ±7.0	1.757 ±1.740	0.325	–
post_proc	100%	0%	0%	79.0 ±21.0	24.5 ±22.5	0.473 ±0.469	0.857	–

post_proc-tight	100%	100%	100%	63.5 ±2.5	0.0 ±0.0	0.000 ±0.000	0.940	–
model_free	100%	50%	50%	69.0 ±8.0	7.0 ±7.0	1.441 ±1.441	0.421	–
model_free-tight	100%	50%	50%	68.5 ±8.5	6.0 ±6.0	1.375 ±1.375	0.431	–
dpcc-c-tight-dt0.25	100%	50%	50%	79.5 ±0.5	1.0 ±1.0	0.013 ±0.011	1.595	–
dpcc-c-tight-dt0.5	100%	100%	100%	65.5 ±1.5	0.0 ±0.0	0.001 ±0.000	1.320	–
dpcc-c-tight-dt2.0	100%	100%	100%	67.0 ±9.0	0.0 ±0.0	0.000 ±0.000	0.766	–
dpcc-c-tight-dt4.0	100%	100%	100%	78.5 ±8.5	0.0 ±0.0	0.000 ±0.000	0.601	–

Section 3 — Head-to-Head Comparison Tables

Each row compares Diffusion (A) vs FM (B) for the same method and scenario. Blue background = A wins, orange = B wins, gray = tie or same value. Winner column shows which model is superior for G&C; (goal+constraint success).

Top-Right-Hard

Method	A G&C;	B G&C;	Winner	A Viol#	B Viol#	A Steps	B Steps	A Time	B Time
dpcc-r	100%	0%	A	0.0	0.5	66.0	0.0	0.698	0.424
dpcc-r-tight	100%	50%	A	0.0	0.0	69.5	62.0	0.817	0.469
dpcc-c	100%	0%	A	0.0	2.0	62.0	71.0	0.744	0.547
dpcc-c-tight	100%	100%	Tie	0.0	0.0	66.5	64.0	0.911	0.585
dpcc-t	50%	0%	A	0.5	1.0	59.0	0.0	0.799	0.537
dpcc-t-tight	100%	100%	Tie	0.0	0.0	61.0	63.5	0.871	0.615
diffuser	100%	0%	A	0.0	33.0	59.0	67.0	0.298	0.303
gradient	100%	0%	A	0.0	34.0	59.0	69.0	0.316	0.321
gradient-tight	0%	0%	Tie	15.0	32.0	58.0	69.0	0.327	0.329
post_proc	100%	0%	A	0.0	0.5	66.0	0.0	0.670	0.444
post_proc-tight	100%	50%	A	0.0	0.0	69.5	62.0	0.814	0.478
model_free	0%	0%	Tie	3.0	35.5	63.5	69.5	0.490	0.409
model_free-tight	50%	0%	A	2.0	35.0	64.0	70.0	0.508	0.416
dpcc-c-tight-dt0.25	100%	100%	Tie	0.0	0.0	81.0	82.0	2.225	1.644
dpcc-c-tight-dt0.5	0%	100%	B	1.0	0.0	75.5	74.5	1.668	1.163
dpcc-c-tight-dt2.0	100%	100%	Tie	0.0	0.0	66.5	69.5	0.689	0.500
dpcc-c-tight-dt4.0	50%	0%	A	0.0	0.0	72.0	0.0	0.535	0.491

Top-Left-Hard

Method	A G&C;	B G&C;	Winner	A Viol#	B Viol#	A Steps	B Steps	A Time	B Time
dpcc-r	100%	50%	A	0.0	7.0	64.5	92.5	0.784	0.753
dpcc-r-tight	100%	50%	A	0.0	34.0	65.5	111.5	0.998	4.675
dpcc-c	50%	50%	Tie	1.0	15.0	62.5	83.0	0.592	0.656
dpcc-c-tight	100%	100%	Tie	0.0	0.0	63.5	62.0	0.822	0.646
dpcc-t	50%	100%	B	1.5	0.0	61.5	62.0	0.829	0.548
dpcc-t-tight	100%	100%	Tie	0.0	0.0	60.0	62.0	0.976	0.614
diffuser	0%	0%	Tie	20.0	11.5	59.0	67.0	0.306	0.304
gradient	0%	0%	Tie	17.5	12.0	58.5	68.5	0.334	0.326
gradient-tight	0%	0%	Tie	19.5	12.5	59.0	68.5	0.348	0.324
post_proc	100%	50%	A	0.0	7.0	64.5	92.5	0.782	0.744
post_proc-tight	100%	50%	A	0.0	34.0	65.5	111.5	0.996	4.615
model_free	0%	0%	Tie	19.0	11.5	63.5	68.5	0.458	0.422

model_free-tight	0%	0%	Tie	15.0	12.0	64.5	69.5	0.470	0.441
dpcc-c-tight-dt0.25	100%	50%	A	0.0	1.0	70.0	89.0	1.259	1.071
dpcc-c-tight-dt0.5	100%	100%	Tie	0.0	0.0	63.5	66.5	0.937	0.563
dpcc-c-tight-dt2.0	100%	100%	Tie	0.0	0.0	64.5	68.0	0.679	0.563
dpcc-c-tight-dt4.0	0%	100%	B	16.5	0.0	107.0	120.0	0.810	0.486

Both-Hard

Method	A G&C;	B G&C;	Winner	A Viol#	B Viol#	A Steps	B Steps	A Time	B Time
dpcc-r	0%	0%	Tie	2.5	24.5	61.5	79.0	1.019	0.859
dpcc-r-tight	100%	100%	Tie	0.0	0.0	66.0	63.5	0.948	0.916
dpcc-c	0%	0%	Tie	19.0	26.5	84.5	90.0	0.704	0.751
dpcc-c-tight	100%	100%	Tie	0.0	0.0	56.5	72.0	0.997	0.779
dpcc-t	50%	0%	A	17.0	29.0	81.5	121.0	0.697	0.573
dpcc-t-tight	100%	100%	Tie	0.0	0.0	62.5	65.5	0.964	0.830
diffuser	0%	50%	B	7.5	8.0	59.0	67.0	0.307	0.300
gradient	0%	50%	B	7.0	7.0	59.0	67.0	0.331	0.326
gradient-tight	0%	50%	B	6.0	7.0	61.0	66.5	0.332	0.325
post_proc	0%	0%	Tie	2.5	24.5	61.5	79.0	1.029	0.857
post_proc-tight	100%	100%	Tie	0.0	0.0	66.0	63.5	0.924	0.940
model_free	0%	50%	B	8.5	7.0	63.0	69.0	0.465	0.421
model_free-tight	0%	50%	B	8.5	6.0	63.5	68.5	0.481	0.431
dpcc-c-tight-dt0.25	50%	50%	Tie	0.5	1.0	75.5	79.5	1.835	1.595
dpcc-c-tight-dt0.5	100%	100%	Tie	0.0	0.0	68.5	65.5	1.185	1.320
dpcc-c-tight-dt2.0	100%	100%	Tie	0.0	0.0	64.5	67.0	0.737	0.766
dpcc-c-tight-dt4.0	100%	100%	Tie	0.0	0.0	85.0	78.5	0.691	0.601

Section 4 — dt Sensitivity: Complete Data

Full breakdown of the dpcc-c-tightened ablation across $dt \in \{0.25, 0.5, 1.0, 2.0, 4.0\}$ for both models and all three scenarios. $dt=1.0$ is the default (same as dpcc-c-tight rows above).

Top-Right-Hard

dt	A G&C;	B G&C;	A Steps $\pm$ std	B Steps $\pm$ std	A Viol# $\pm$ std	B Viol# $\pm$ std	A Viol Σ $\pm$ std	B Viol Σ $\pm$ std	A Time	B Time
dt=0.25	100%	100%	81.0 $\pm$ 6.0	82.0 $\pm$ 5.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.002 $\pm$ 0.000	0.003 $\pm$ 0.002	2.225	1.644
dt=0.5	0%	100%	75.5 $\pm$ 2.5	74.5 $\pm$ 6.5	1.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.003 $\pm$ 0.000	0.001 $\pm$ 0.001	1.668	1.163
dt=1.0 (default)	100%	100%	66.5 $\pm$ 0.5	64.0 $\pm$ 6.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.911	0.585
dt=2.0	100%	100%	66.5 $\pm$ 0.5	69.5 $\pm$ 8.5	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.689	0.500
dt=4.0	50%	0%	72.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.535	0.491

Top-Left-Hard

dt	A G&C;	B G&C;	A Steps $\pm$ std	B Steps $\pm$ std	A Viol# $\pm$ std	B Viol# $\pm$ std	A Viol Σ $\pm$ std	B Viol Σ $\pm$ std	A Time	B Time
dt=0.25	100%	50%	70.0 $\pm$ 3.0	89.0 $\pm$ 7.0	0.0 $\pm$ 0.0	1.0 $\pm$ 1.0	0.000 $\pm$ 0.000	0.013 $\pm$ 0.009	1.259	1.071
dt=0.5	100%	100%	63.5 $\pm$ 0.5	66.5 $\pm$ 0.5	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.937	0.563
dt=1.0 (default)	100%	100%	63.5 $\pm$ 3.5	62.0 $\pm$ 3.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.822	0.646
dt=2.0	100%	100%	64.5 $\pm$ 0.5	68.0 $\pm$ 8.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.679	0.563
dt=4.0	0%	100%	107.0 $\pm$ 2.0	120.0 $\pm$ 6.0	16.5 $\pm$ 0.5	0.0 $\pm$ 0.0	3.711 $\pm$ 0.053	0.000 $\pm$ 0.000	0.810	0.486

Both-Hard

dt	A G&C;	B G&C;	A Steps $\pm$ std	B Steps $\pm$ std	A Viol# $\pm$ std	B Viol# $\pm$ std	A Viol Σ $\pm$ std	B Viol Σ $\pm$ std	A Time	B Time
dt=0.25	50%	50%	75.5 $\pm$ 0.5	79.5 $\pm$ 0.5	0.5 $\pm$ 0.5	1.0 $\pm$ 1.0	0.000 $\pm$ 0.000	0.013 $\pm$ 0.011	1.835	1.595
dt=0.5	100%	100%	68.5 $\pm$ 4.5	65.5 $\pm$ 1.5	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.001 $\pm$ 0.000	1.185	1.320
dt=1.0 (default)	100%	100%	56.5 $\pm$ 2.5	72.0 $\pm$ 12.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.997	0.779
dt=2.0	100%	100%	64.5 $\pm$ 2.5	67.0 $\pm$ 9.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.737	0.766
dt=4.0	100%	100%	85.0 $\pm$ 2.0	78.5 $\pm$ 8.5	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	0.691	0.601

Section 5 — Summary Statistics & Key Findings

5.1 Aggregate Statistics — Core Methods (13 methods × 3 scenarios = 39 observations each)

Metric	Diffusion (A) mean±std	FM (B) mean±std	Delta (A–B)
Goal & Constraint Success	0.551 ± 0.464	0.385 ± 0.400	+0.167
Constraint Satisfaction	0.551 ± 0.464	0.385 ± 0.400	+0.167
Task Success Rate	0.987 ± 0.079	0.885 ± 0.288	+0.103
Avg Violations / Episode	4.936 ± 7.002	12.038 ± 12.852	-7.103
Avg Total Violation	0.372 ± 0.803	1.207 ± 2.285	-0.836
Avg Compute Time (s/step)	0.671 ± 0.251	0.750 ± 0.925	-0.079

5.2 Win / Tie / Loss Count — G&C; Success Rate

Head-to-head wins across all core method × scenario combinations (39 total).

Diffusion (A) wins: 14	Ties: 19	FM (B) wins: 6
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5.3 Key Findings

1. Diffusion is the stronger base model

Diffusion wins 14/39 non-tied comparisons on G&C; success rate. Its mean G&C; success is 16.7pp higher than FM across all core methods and scenarios.

2. FM collapses on top-right-hard without tightening

dpcc-r, dpcc-c (untightened), dpcc-t, post_proc, and model_free all score 0% G&C; for FM on top-right-hard. Diffusion achieves 0–100% on the same methods. FM's diffuser/gradient baselines produce 32–35 violations/episode.

3. dpcc-c-tightened is universally reliable

Both models achieve 100% G&C; on dpcc-c-tightened across all three scenarios. This is the safest method choice regardless of base model.

4. Tightening is essential for FM, optional for Diffusion

Without tightening, FM frequently fails even when Diffusion succeeds. Constraint projection tightening compensates for FM's weaker constraint awareness.

5. FM has one notable advantage: top-left dpcc-t

FM's dpcc-t (untightened) achieves 100% on top-left-hard vs only 50% for Diffusion, suggesting FM trajectories are better suited to left-side obstacle avoidance in this configuration.

6. dt sensitivity: FM more fragile at extremes

At dt=4.0, FM fails entirely on top-right-hard (0 steps taken). At dt=0.5, Diffusion fails on top-right-hard (0%). Both are most robust at dt=1.0–2.0. FM's dpcc-r-tightened at dt=4.0 (top-left) is extreme: 4.675 s/step and highly variable steps.

7. Computation time: Diffusion is slower on DPCC, faster on baselines

Diffusion's dpcc-c-tight averages ~0.91s/step vs FM's ~0.67s/step. However Diffusion's diffuser/gradient baselines are similar to FM (~0.30s). FM's dpcc-r-tightened on top-left at dt=0.5 reaches 4.675s/step — an outlier likely due to SLSQP clipping warnings.

8. Both models have 100% task success rate on all methods

The success_rate metric (reaching goal regardless of constraints) is 1.0 for nearly all methods. The discriminating metric is goal+constraints combined.

Data source: avoiding-d3il benchmark, H8_K20_Dmodels, seed 6. A = GaussianDiffusion (diffusion branch). B = GaussianDiffusion in flow_matching branch. † = tightened constraint projection variant. All step counts and violation counts are averaged over 2 seeds per episode pair.